

KAHROBAEI-KOUPPARIS DSS: UNIVERSAL FORGERY

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ABSTRACT. Regardless of the choice of parameters, knowledge of a single signed message, i.e., a pair message/signature, produced by Kahrobaei-Koupparis digital signature scheme, proposed in [1], is sufficient to forge a valid signature for any other message.

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1. THE KAHROBAEI-KOUPPARIS DSS SCHEME

Let G be a group. Fix two functions:

- $f : G \rightarrow \{0, 1\}^*$, an encoding function that maps group elements to binary strings;
- $H : \{0, 1\}^* \rightarrow G$, a collision-resistant hash function mapping binary strings to elements of G .

The Kahrobaei-Koupparis DSS scheme digital signature scheme consists of the following steps.

- (Key-generation) The signer first chooses
 - an element $g \in G$, whose centralizer $C(g)$ is the cyclic subgroup $\langle g \rangle$,
 - $s \in G$,
 - $n \in N$, where n is chosen to be highly composite.

The private key is a pair (s, n) . The public key is the element $x = g^{ns} \in G$.

- (Signing) To sign a message m , the signer chooses a random element $t \in G$ and a random factorization $n_i n_j$ of n , and computes the following (with $\parallel$ denoting concatenation):
 - $y = g^{n_i t}$,
 - $h = H(m \parallel f(y))$,
 - $\alpha = t^{-1} s h y$.

Then $\text{sign}(m) = (y, \alpha, n_j)$.

- (Verification) A given signature (y, α, n_j) for m is verified by
 - computing $h' = H(m \parallel f(y))$
 - verifying the identity $y^{n_j \alpha} = x^{h' y}$.

Proposition 1.1. *The scheme is correct.*

Proof. Indeed a signature constructed according to the protocol yields $h' = h$ during verification, which then results in equal elements

- $y^{n_j \alpha} = (g^{n_i t})^{n_j \alpha} = (g^{nt})^\alpha = g^{ntt^{-1} s h y} = g^{n s h y}$,
- $x^{h y} = (g^{n s})^{h y} = g^{n s h y}$.

□

2. FORGING SIGNATURES

Let (y, α, n_j) be a signature for m . Then we can compute $h = H(m \| f(y))$ and, hence,

$$\Delta = \alpha y^{-1} h^{-1},$$

which is obviously the same as $t^{-1}s$. Then for a message m' we can

- keep the same value y ,
- compute $h' = H(m' \| f(y))$,
- compute $\alpha' = \Delta h' y$.

Form the tuple (y, α', n_j) .

Proposition 2.1. *(y, α', n_j) passes the verification step of the scheme as a signature for m' .*

Proof. Indeed, during verification the value $h' = H(m' \| f(y))$ is computed. Then we can directly check that

- $y^{n_j \alpha'} = (g^{n_i t})^{n_j \alpha'} = (g^{nt})^{\alpha'} = g^{nt \Delta h' y} = g^{nt t^{-1} s h' y} = g^{n s h' y}$,
- $x^{h' y} = (g^{ns})^{h' y} = g^{n s h' y}$.

□

REFERENCES

- [1] D. Kahrobaei and C. Koupparis. Non-commutative digital signatures using non-commutative groups. *Groups, Complexity, Cryptology*, 4, 2012.

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